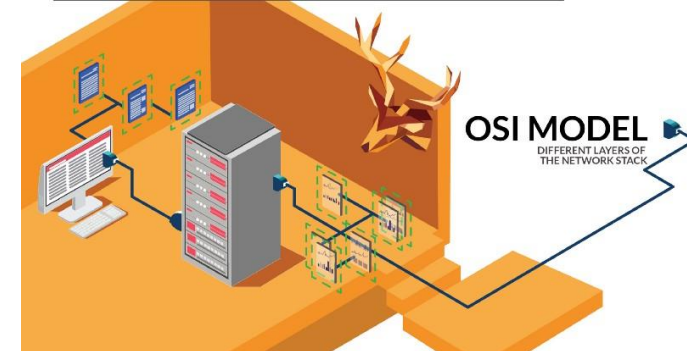




HTTP, Wireshark, and Digital Forensics

Part 2: HTTP traffic contains embedded images

Traffic files:

<https://1drv.ms/f/c/a674908a5e2e2f2a/IgDzyvW3yHkdSIWBdVAk82DLAafpQ6YrXyVbQ7S0VAAd73Y?e=iWXBjV>



Overview

- Capture HTTP traffic (with a picture embedded)
- Captured HTTP Traffic Overview
- Analyze the second HTTP get request/response
 - Traffic contains the picture
- Extract the building.jpg from the traffic
- Size relation between IP packet and TCP payload



Capture HTTP traffic

with a picture embedded



- Download an image **building.jpg**
- Save it to **/var/www/html**

```
root@kali:~#  
root@kali:~# wget -q https://www.dropbox.com/s/jiq17tgi8u1xls3/building.jpg -P /var/www/html/  
root@kali:~#  
root@kali:~# ls -l /var/www/html/building.jpg  
-rw-r--r-- 1 root root 68661 Oct 26 16:40 /var/www/html/building.jpg
```

-P: directory prefix

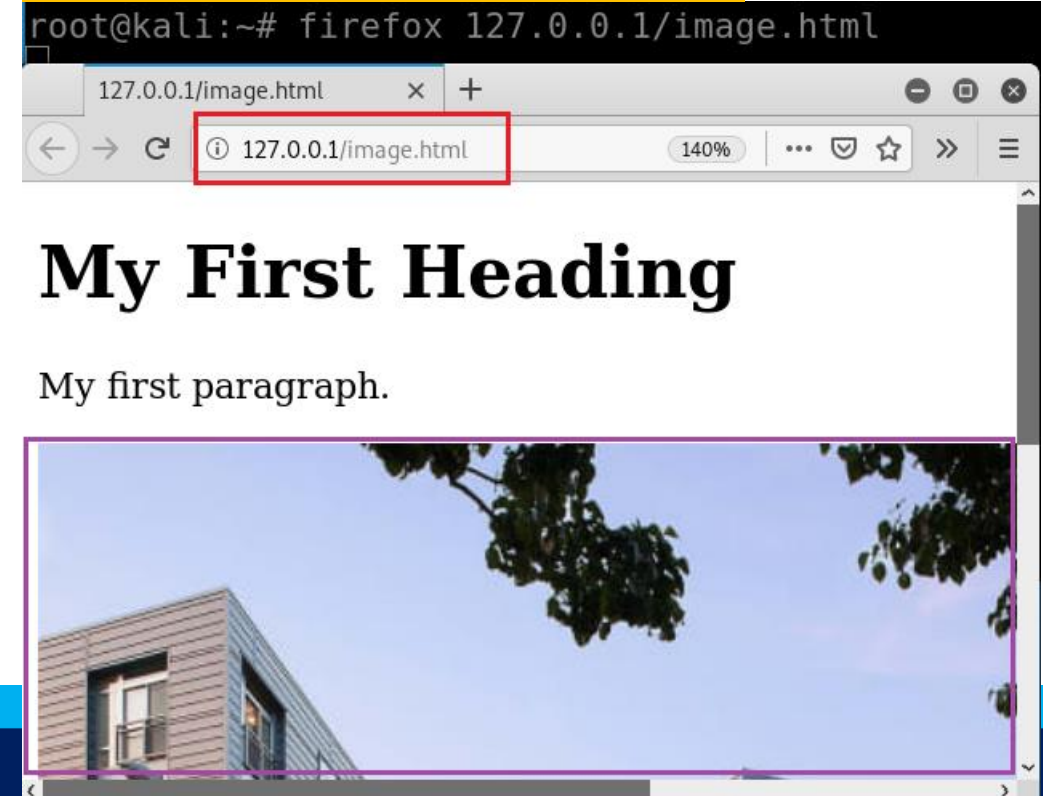
Add **building.jpg** in a webpage **image.html**

```
root@kali:~# leafpad /var/www/html/image.html
```



```
<!DOCTYPE html>  
<html>  
<body>  
  
<h1>My First Heading</h1>  
  
<p>My first paragraph.</p>  
  
  
  
</body>  
</html>
```

Show the webpage with the image



https://raw.githubusercontent.com/frankwxu/digital-forensics-lab/main/Illegal_Possession_Images/lab_files/traffic/image.html



Capture traffic

```
/bin/bash 85x7
root@kali:~# rm -r ~/.cache/mozilla/firefox/*default* 1
rm: cannot remove '/root/.cache/mozilla/firefox/*default*': No such file or directory
root@kali:~# wireshark -i lo -k -w ~/traffic/image2.log 2
Warning: Ignoring XDG_SESSION_TYPE=wayland on Gnome. Use QT_QPA_PLATFORM=wayland to run on Wayland anyway.
^C ← 5
root@kali:~#

/bin/bash 85x14
root@kali:~# firefox 127.0.0.1/image.html 3
^C ← 4
Exiting due to channel error.
Exiting due to channel error.
Exiting due to channel error.
Exiting due to channel error.

root@kali:~#
```

HTTP Traffic overview



How many http **get** requests and responses?

```
root@kali:~# leafpad /var/www/html/image.html
```

image.html

File Edit Search Options Help

<!DOCTYPE html>

<html>

<body>

<h1>My First Heading</h1>

<p>My first paragraph.</p>

</body>

</html>



Port 52054



Port 80

GET 127.0.0.1/image.html



Parse image.html

```

File Edit Search Options Help
<!DOCTYPE html>
<html>
<body>

<h1>My First Heading</h1>

<p>My first paragraph.</p>



</body>
</html>

```

```

root@kali:~#
root@kali:~# ls -l /var/www/html/image.html
-rw-r--r-- 1 root root 139 Oct 26 16:05 /var/www/html/image.html
root@kali:~#

```

GET 127.0.0.1/building.jpg

32768 bytes : default value of for TCP send and receive buffers



32768 bytes

```

/bin/bash 85x7
root@kali:~#
root@kali:~# ls -l /var/www/html/building.jpg
-rw-r--r-- 1 root root 68661 Oct 26 16:40 /var/www/html/building.jpg
root@kali:~#

```



32768 bytes



3414 bytes

http response head length (289) + Build.jpg (68661) = TCP 1(32768) + TCP 2(32768) + TCP 3 (3414) = 68950





wget https://raw.githubusercontent.com/frankwxu/digital-forensics-lab/main/Illegal_Possession_Images/lab_files/traffic/image2.log

Verify two HTTP GET: .html and .jpg

```
root@kali:~#
root@kali:~# wireshark traffic/image2.log
```

image2.log

File Edit View Go Capture Analyze Statistics Telephony Wireless Tools Help

Apply a display filter ... <Ctrl-/>

No.	Time	Source	Destination	Protocol	Length	Info
1	0.000000000	127.0.0.1	127.0.0.1	TCP	74	52054 → 80 [SYN] Seq=0 Win=65495 Len=0 MSS=65495 SA
2	0.0000008840	127.0.0.1	127.0.0.1	TCP	74	80 → 52054 [SYN, ACK] Seq=0 Ack=1 Win=65483 Len=0 M
3	0.0000015234	127.0.0.1	127.0.0.1	TCP	66	52054 → 80 [ACK] Seq=1 Ack=1 Win=65536 Len=0 TSval=
4	0.386287395	127.0.0.1	127.0.0.1	HTTP	385	GET /image.html HTTP/1.1
5	0.386311944	127.0.0.1	127.0.0.1	TCP	66	80 → 52054 [ACK] Seq=1 Ack=320 Win=65280 Len=0 TSva
6	0.386567159	127.0.0.1	127.0.0.1	HTTP	534	HTTP/1.1 200 OK (text/html)
7	0.386576991	127.0.0.1	127.0.0.1	TCP	66	52054 → 80 [ACK] Seq=320 Ack=469 Win=65152 Len=0 TS
8	0.716114099	127.0.0.1	127.0.0.1	HTTP	346	GET /building.jpg HTTP/1.1
9	0.716125972	127.0.0.1	127.0.0.1	TCP	66	80 → 52054 [ACK] Seq=469 Ack=600 Win=65280 Len=0 TS
10	0.716341907	127.0.0.1	127.0.0.1	TCP	328...	80 → 52054 [ACK] Seq=469 Ack=600 Win=65536 Len=3276
11	0.716353387	127.0.0.1	127.0.0.1	TCP	66	52054 → 80 [ACK] Seq=600 Ack=33237 Win=48512 Len=0
12	0.716363658	127.0.0.1	127.0.0.1	TCP	328...	80 → 52054 [PSH, ACK] Seq=33237 Ack=600 Win=65536
13	0.716368424	127.0.0.1	127.0.0.1	TCP	66	52054 → 80 [ACK] Seq=600 Ack=66005 Win=15744 Len=0
14	0.716376090	127.0.0.1	127.0.0.1	HTTP	3480	HTTP/1.1 200 OK (JPEG JFIF image)
15	0.716388599	127.0.0.1	127.0.0.1	TCP	66	[TCP Window Update] 52054 → 80 [ACK] Seq=600 Ack=66
16	0.716445359	127.0.0.1	127.0.0.1	TCP	66	52054 → 80 [ACK] Seq=600 Ack=69419 Win=65536 Len=0
17	0.863058108	127.0.0.1	127.0.0.1	HTTP	307	GET /favicon.ico HTTP/1.1
18	0.863063656	127.0.0.1	127.0.0.1	TCP	66	80 → 52054 [ACK] Seq=69419 Ack=841 Win=65408 Len=0
19	0.863194653	127.0.0.1	127.0.0.1	HTTP	553	HTTP/1.1 404 Not Found (text/html)
20	0.863242007	127.0.0.1	127.0.0.1	TCP	66	52054 → 80 [ACK] Seq=841 Ack=69906 Win=65536 Len=0
21	3.982090205	127.0.0.1	127.0.0.1	TCP	66	52054 → 80 [FIN, ACK] Seq=841 Ack=69906 Win=65536 L
22	3.984114989	127.0.0.1	127.0.0.1	TCP	66	80 → 52054 [FIN, ACK] Seq=69906 Ack=842 Win=65536 L
23	3.984119529	127.0.0.1	127.0.0.1	TCP	66	52054 → 80 [ACK] Seq=842 Ack=69907 Win=65536 Len=0

First HTTP GET html

Second HTTP GET .jpg

Third HTTP GET .ico



Analyzing the second HTTP get request/response

HTTP traffic for displaying images



8	0.716114099	127.0.0.1	127.0.0.1	HTTP	346 GET /building.jpg HTTP/1.1
9	0.716125972	127.0.0.1	127.0.0.1	TCP	66 80 → 52054 [ACK] Seq=469 Ack=600 Win=65280 Len=0 TS
10	0.716341907	127.0.0.1	127.0.0.1	TCP	328... 80 → 52054 [ACK] Seq=469 Ack=600 Win=65536 Len=32768
11	0.716353387	127.0.0.1	127.0.0.1	TCP	66 52054 → 80 [ACK] Seq=600 Ack=33237 Win=48512 Len=0
12	0.716363658	127.0.0.1	127.0.0.1	TCP	328... 80 → 52054 [PSH, ACK] Seq=33237 Ack=600 Win=65536 Len=32768
13	0.716368424	127.0.0.1	127.0.0.1	TCP	66 52054 → 80 [ACK] Seq=600 Ack=66005 Win=15744 Len=0
14	0.716376090	127.0.0.1	127.0.0.1	HTTP	3480 HTTP/1.1 200 OK (JPEG JFIF image)
15	0.716388599	127.0.0.1	127.0.0.1	TCP	66 [TCP Window Update] 52054 → 80 [ACK] Seq=600 Ack=66005 Win=65536 Len=0
16	0.716445359	127.0.0.1	127.0.0.1	TCP	66 52054 → 80 [ACK] Seq=600 Ack=69419 Win=65536 Len=0

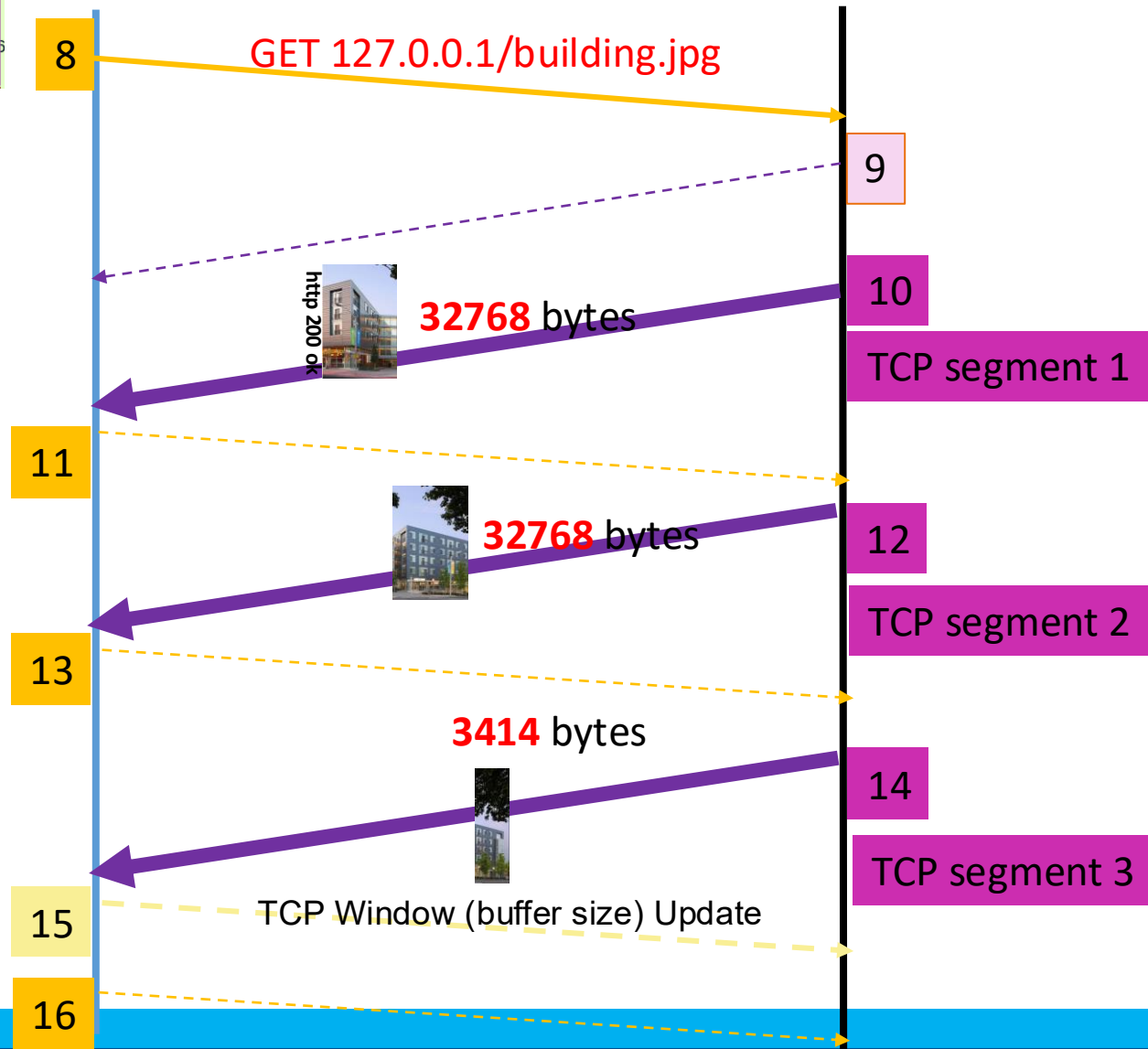
Second HTTP GET.jpg



Port 52054



Port 80



- TCP segment reassembly takes places in HTTP layer.
- TCP reassembling at receiving side
- Wireshark shows the assembled TCP segment at sending side
- TCP segment reassembly works only capture the entire packet and if all the checksums for that packet are valid.

Calculate the length 3 TCP segments: 32768+32768+3414

No.	Time	Source	Destination	Protocol	Length	Info
9	0.716125972	127.0.0.1	127.0.0.1	TCP	66	80 → 52054 [ACK] Seq=469 Ack=600 Win=65280 Len=0 TSval=11185
10	0.716341907	127.0.0.1	127.0.0.1	TCP	328...	80 → 52054 [ACK] Seq=469 Ack=600 Win=65536 Len=32768 TSval=1
11	0.716353387	127.0.0.1	127.0.0.1	TCP	66	52054 → 80 [ACK] Seq=600 Ack=33237 Win=48512 Len=0 TSval=111
12	0.716363658	127.0.0.1	127.0.0.1	TCP	328...	80 → 52054 [PSH, ACK] Seq=33237 Ack=600 Win=65536 Len=32768
13	0.716368424	127.0.0.1	127.0.0.1	TCP	66	52054 → 80 [ACK] Seq=600 Ack=66005 Win=15744 Len=0 TSval=111
14	0.716376090	127.0.0.1	127.0.0.1	HTTP	3480	HTTP/1.1 200 OK (JPEG JFIF image)
15	0.716388599	127.0.0.1	127.0.0.1	TCP	66	[TCP Window Update] 52054 → 80 [ACK] Seq=600 Ack=66005 Win=4
16	0.716445359	127.0.0.1	127.0.0.1	TCP	66	52054 → 80 [ACK] Seq=600 Ack=69419 Win=65536 Len=0 TSval=111

Sequence number: 66005 (relative sequence number)
Sequence number (raw): 1720224929
[Next sequence number: 69419 (relative sequence number)]
Acknowledgment number: 600 (relative ack number)
Acknowledgment number (raw): 1950579563
1000 = Header Length: 32 bytes (8)
► Flags: 0x018 (PSH, ACK)
Window size value: 512
[Calculated window size: 65536]
[Window size scaling factor: 128]
Checksum: 0x0b7f [unverified]
[Checksum Status: Unverified]
Urgent pointer: 0
► Options: (12 bytes), No-Operation (NOP), No-Operation (NOP), Timestamps
► [SEQ/ACK analysis]
► [Timestamps]
TCP payload (3414 bytes)
TCP segment data (3414 bytes)
[3 Reassembled TCP Segments (68950 bytes): #10(32768), #12(32768), #14(3414)]
► Hypertext Transfer Protocol

00000000	48 54 54 50 2f 31 2e 31 20 32 30 30 20 4f 4b 0d	HTTP/1.1 200 OK.
00000010	0a 44 61 74 65 3a 20 53 75 6e 2c 20 30 38 20 4e	Date: Sun, 08 N

TCP segments



Show the content of 3 TCP segments is http response and the data (building.jpg)

14

14 0.716376090 127.0.0.1 127.0.0.1 HTTP 3480 HTTP/1.1 200 OK (JPEG JFIF image) 7

15 0.716388599 127.0.0.1 127.0.0.1 TCP 66 [TCP Window Update] 52054 → 80 [ACK] Seq=600 Ack=66005

16 0.716445359 127.0.0.1 127.0.0.1 TCP 66 52054 → 80 [ACK] Seq=600 Ack=69419 Win=65536 Len=0 TSva

17 0.863058108 127.0.0.1 127.0.0.1 HTTP 307 GET /favicon.ico HTTP/1.1

- ▶ Frame 14: 3480 bytes on wire (27840 bits), 3480 bytes captured (27840 bits) on interface lo, id 0
- ▶ Ethernet II, Src: 00:00:00_00:00:00 (00:00:00:00:00:00), Dst: 00:00:00_00:00:00 (00:00:00:00:00:00)
- ▶ Internet Protocol Version 4, Src: 127.0.0.1, Dst: 127.0.0.1
- ▶ Transmission Control Protocol, Src Port: 80, Dst Port: 52054, Seq: 66005, Ack: 600, Len: 3414
- ▶ [3 Reassembled TCP Segments (68950 bytes): #10(32768), #12(32768), #14(3414)] 2
- ▶ Hypertext Transfer Protocol
- ▶ JPEG File Interchange Format

00000000 48 54 54 50 2f 31 2e 31 20 32 30 30 20 4f 4b 0d HTTP/1.1 200 OK.
00000010 0a 44 61 74 65 3a 20 53 75 6e 2c 20 30 38 20 4e .Date: Sun, 08 N
00000020 6f 76 20 32 30 32 30 20 31 35 3a 30 35 3a 30 38 ov 2020 15:05:08
00000030 20 47 4d 54 0d 0a 53 65 72 76 65 72 3a 20 41 70 GMT..Se rver: Ap
00000040 61 63 68 65 2f 32 2e 34 2e 34 31 20 28 44 65 62 ache/2.4 .41 (Deb
00000050 69 61 6e 29 0d 0a 4c 61 73 74 2d 4d 6f 64 69 66 ian)..La st-Modif
00000060 69 65 64 3a 20 4d 6f 6e 2c 20 32 36 20 4f 63 74 ied: Mon , 26 Oct
00000070 20 32 30 32 30 20 32 30 3a 34 30 3a 31 37 20 47 2020 20 :40:17 G
00000080 4d 54 0d 0a 45 54 61 67 3a 20 22 31 30 63 33 35 MT..ETag : "10c35
00000090 2d 35 62 32 39 38 66 30 32 35 62 35 34 34 22 0d -5b298f0 25b544".
000000a0 0a 41 63 63 65 70 74 2d 52 61 6e 67 65 73 3a 20 .Accept- Ranges:
000000b0 62 79 74 65 73 0d 0a 43 6f 6e 74 65 6e 74 2d 4c bytes..C ontent-L
000000c0 65 6e 67 74 68 3a 20 36 38 36 36 31 0d 0a 4b 65 length: 6 8661..Ke
000000d0 65 70 2d 41 6c 69 76 65 3a 20 74 69 6d 65 6f 75 ep-Alive : timeou
000000e0 74 3d 35 2c 20 6d 61 78 3d 39 39 0d 0a 43 6f 6e t=5, max =99..Con
000000f0 6e 65 63 74 69 6f 6e 3a 20 4b 65 65 70 2d 41 6c nnection. Keep Al
00000100 69 76 65 0d 0a 43 6f 6e 74 65 6e 74 2d 54 79 70 ive. Con tent-Typ
00000110 65 3a 20 69 6d 61 67 65 2f 6a 70 65 67 0d 0a 0d e: image/jpeg...

Frame (3480 bytes) Reassembled TCP (68950 bytes)

3. 3 Reassembled TCP segments are HTTP response. HTTP response contains HTTP response header and image

Calculate the length of http response header: 289 (0x000 -0x120)

14

14 0.716376090 127.0.0.1 127.0.0.1 HTTP 3480 HTTP/1.1 200 OK (JPEG JFIF image)

Hypertext Transfer Protocol

HTTP/1.1 200 OK\r\n

Date: Sun, 08 Nov 2020 15:05:08 GMT\r\n

Server: Apache/2.4.41 (Debian)\r\n

Last-Modified: Mon, 26 Oct 2020 20:40:17 GMT\r\n

ETag: "10c35-5b298f025b544"\r\n

Accept-Ranges: bytes\r\n

Content-Length: 68661\r\n

Keep-Alive: timeout=5, max=99\r\n

Connection: Keep-Alive\r\n

Content-Type: image/jpeg\r\n

\r\n

0d0a

289 byte

00000000 48 54 54 50 2f 31 2e 31 20 37 30 30 20 4f 4b 0d HTTP/1.1 200 OK
00000010 0a 44 61 74 65 3a 20 53 75 3e 2c 20 30 38 20 4e Date: Sun, 08 N
00000020 6f 76 20 32 30 32 30 20 31 35 3a 30 35 3a 30 38 ov 2020 15:05:08
00000030 20 47 4d 54 0d 0a 53 65 72 76 65 72 3a 20 41 70 GMT Server: Ap
00000040 61 63 68 65 2f 32 2e 34 2e 34 31 20 28 44 65 62 ache/2.4 .41 (Deb
00000050 69 61 6e 29 0d 0a 4c 61 73 74 2d 4d 6f 64 69 66 ian) Last-Modif
00000060 69 65 64 3a 20 4d 6f 6e 2c 20 32 36 20 4f 63 74 ied: Mon , 26 Oct
00000070 20 32 30 30 30 30 30 30 3a 34 30 3a 31 37 20 47 2020 20 :40:17 G
00000080 4d 54 0d 0a 4c 61 6e 67 73 74 2d 4d 6f 64 69 66 MT ETag : "10c35
00000090 2d 35 62 32 39 38 76 30 32 35 34 34 22 0d -5b298f0 25b544".
000000a0 0a 41 63 63 65 70 74 2d 52 31 6e 67 65 73 3a 20 Accept- Ranges:
000000b0 62 79 74 65 73 0d 0a 43 6f 6e 74 65 6e 74 2d 4c bytes Content-L
000000c0 65 6e 67 74 68 3a 20 36 38 36 36 31 0d 0a 4b 65 length: 6 8661 Ke
000000d0 65 70 2d 41 6c 69 76 65 3a 20 74 69 6d 65 6f 75 ep-Alive : timeou
000000e0 74 3d 35 2c 20 6d 61 78 3d 39 39 0d 0a 43 6f 6e t=5, max =99 Con
000000f0 6e 65 63 74 69 6f 6e 3a 20 4b 65 65 70 2d 41 6c nection: Keep-Al
00000100 69 76 65 0d 0a 43 6f 6e 74 65 6e 74 2d 54 79 70 ive Content-Typ
00000110 65 3a 20 69 6d 61 67 65 2f 6a 70 65 67 0d 0a 0d e: image /jpeg
00000120 0a 4f d8 ff e0 00 10 4a 46 49 46 00 01 01 00 00 J FIF



Verify start and end of Image (0x0121 – 0x10d55 +1 =68661 bytes)

14

```
12 0.716363658 127.0.0.1 127.0.0.1 TCP 328... 80 → 52054 [PSH, ACK] Seq=33237 Ack=600 Win=
13 0.716368424 127.0.0.1 127.0.0.1 TCP 66 52054 → 80 [ACK] Seq=600 Ack=66005 Win=1574
14 0.716376090 127.0.0.1 127.0.0.1 HTTP 3480 HTTP/1.1 200 OK (JPEG JFIF image)
15 0.716388599 127.0.0.1 127.0.0.1 TCP 66 [TCP Window Update] 52054 → 80 [ACK] Seq=600
16 0.716445359 127.0.0.1 127.0.0.1 TCP 66 52054 → 80 [ACK] Seq=600 Ack=69419 Win=6553
```

TCP payload (3414 bytes)

TCP segment data (3414 bytes)

▶ [3 Reassembled TCP Segments (68950 bytes): #10(32768), #12(32768), #14(3414)]

▶ Hypertext Transfer Protocol

▶ JPEG File Interchange Format

Marker: Start of Image (0xffd8) ←

▶ Marker segment: Reserved for application segments - 0 (0xFFE0)

▶ Marker segment: Define quantization table(s) (0xFFDB)

▶ Marker segment: Define quantization table(s) (0xFFDB)

▶ Start of Frame header: Start of Frame (non-differential, Huffman coding) - Baseline DCT (0xFFC0)

▶ Marker segment: Define Huffman table(s) (0xFFC4)

▶ Marker segment: Define Huffman table(s) (0xFFC4)

▶ Marker segment: Define Huffman table(s) (0xFFC4)

▶ Marker segment: Define Huffman table(s) (0xFFC4)

▶ Start of Segment header: Start of Scan (0xFFDA)

Entropy-coded segment (dissection is not yet implemented): f6baec321281094c04c50026280131400628

Marker: End of Image (0xffd9) ←

```
00000120 0a ff d8 ff e0 00 10 4a 46 49 46 00 01 01 00 00
00000130 01 00 01 00 00 ff db 00 43 00 06 06 06 06 07 06
00000140 07 08 08 07 0a 0b 0a 0b 0a 0f 0e 0c 0c 0e 0f 16
00000150 10 11 10 11 10 16 22 15 19 15 15 19 15 22 1e 24
00000160 1e 1c 1e 24 1e 36 2a 26 26 2a 36 3e 34 32 34 3e
00000170 4c 44 44 4c 5f 5a 5f 7c 7c a7 ff db 00 43 01 0d
00000180 0d 0d 0d 0e 0d 0e 10 10 0e 14 16 13 16 14 1e 1b
00000190 19 19 1b 1e 2d 20 22 20 22 20 2d 44 2a 32 2a 2a
```

```
.....J FIF.....
..... C.....
..... ".....". $
...$.6*& &*6>424>
LDDL_Z_| |...C..
.....
..... " " -D*2**
```

Frame (3480 bytes)

Reassembled TCP (68950 bytes)



Extract the building.jpg from traffic



How to extract the building.jpg from log

The image shows the Wireshark network protocol analyzer interface. The packet list on the left shows packet 14 selected, which is an HTTP 200 OK response containing a JPEG image. The packet details pane on the right shows the 'JPEG File Interchange Format' section expanded. The context menu is open over the 'JPEG File Interchange Format' section, and the 'Export Packet Bytes...' option is highlighted. The packet bytes pane at the bottom shows the raw data of the selected packet.

1

2

3

14 0.716376090 127.0.0.1 127.0.0.1 HTTP 3480 HTTP/1.1 200 OK (JPEG JFIF image)

15 0.716388599 127.0.0.1 127.0.0.1 TCP 66 [TCP Window Update] 52054 → 80 [ACK] S

16 0.716445359 127.0.0.1 127.0.0.1 TCP 66 52054 → 80 [ACK] Seq=600 Ack=69419 Win

JPEG File Interchange Format

- Marker: Start of Image (0x)
- ▶ Marker segment: Reserved f
- ▶ Marker segment: Define qua
- ▶ Marker segment: Define qua
- ▶ Start of Frame header: Sta
- ▶ Marker segment: Define Huf
- ▶ Marker segment: Define Huf
- ▶ Marker segment: Define Huf
- ▶ Marker segment: Define Huf
- ▶ Start of Segment header: S
- Entropy-coded segment (dis
- Marker: End of Image (0xff

00010ca0 b2 fb d8 d6 90 3f

00010cb0 07 d1 fd e5 77 89

00010cc0 84 f0 ab bc be e4

00010cd0 d3 34 ef e4 64 e8

00010ce0 c7 e8 41 fe 54 73

00010cf0 a3 d8 1f c6 8e 74

00010d00 bf 51 4f 99 10 e8

00010d10 9c 5a 13 26 80 b0

00010d20 40 06 e1 8e b4 00

00010d30 9f 6a 40 1b bd a8

00010d40 3e 94 00 6e e6 80

00010d50 db de 80 3f ff d9

Expand Subtrees

Collapse Subtrees

Expand All

Collapse All

Apply as Column Ctrl+Shift+I

Apply as Filter

Prepare as Filter

Conversation Filter

Colorize with Filter

Follow

Copy

Show Packet Bytes... Ctrl+Shift+O

Export Packet Bytes... Ctrl+Shift+X

Wiki Protocol Page

Filter Field Reference

Protocol Preferences

Decode As...

Go to Linked Packet

Show Linked Packet in New Window

an coding) - Baseline DCT (C

aec321281094c04c500262801314

..?D. .g..G0..

.w..~ i.....9

..St U.....]

.d.G. .C.?,.

A.Tsy. P.....;

.t?. V...q.X

0...T[.,....~

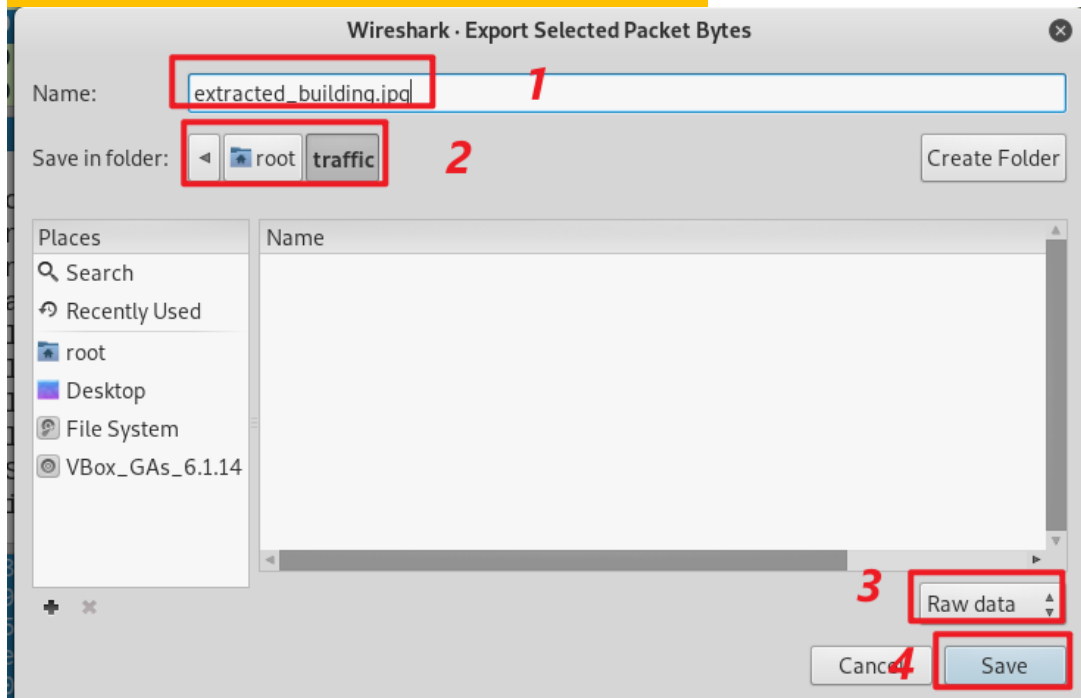
&.... @.a.

...4 ...!...

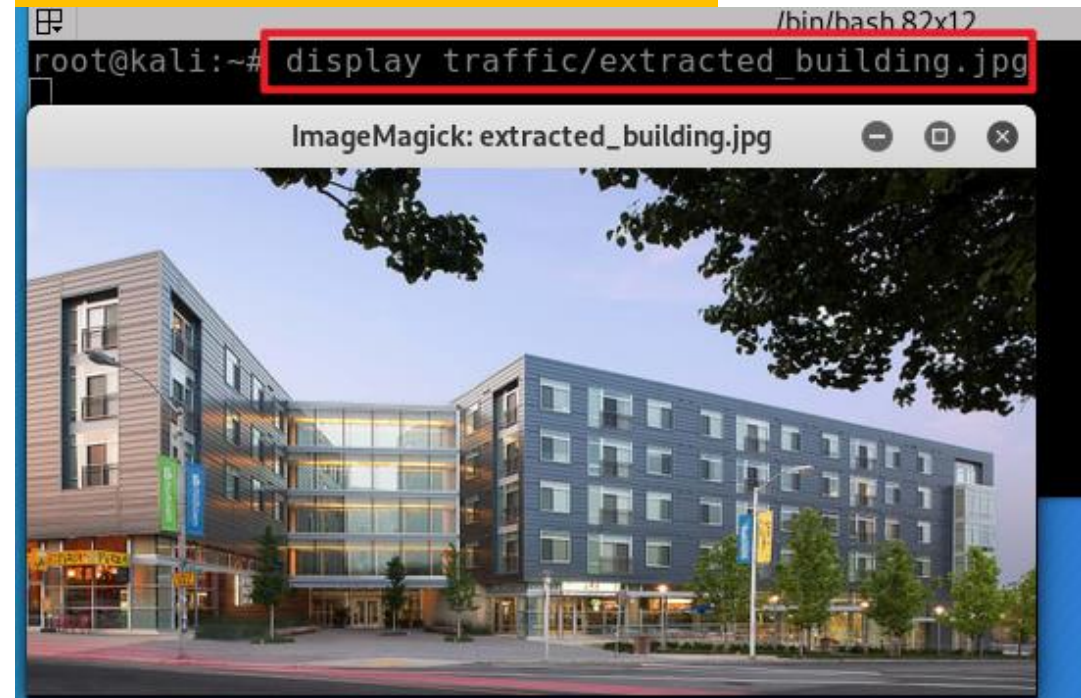
.....(

n...0 ..y...a

Save extracted image

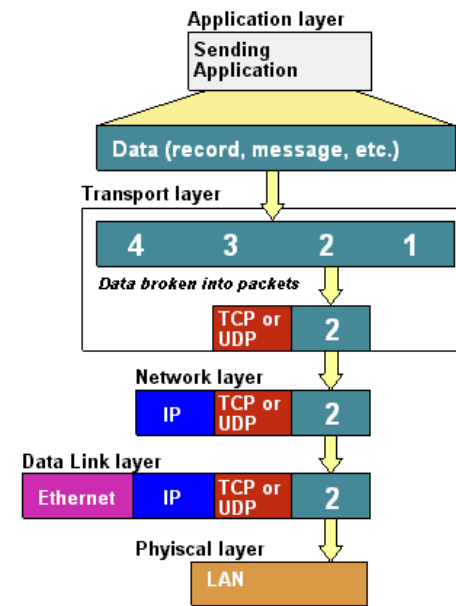
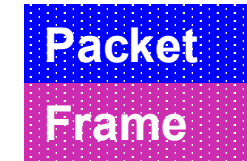


Show image



```
/bin/bash 79x10
root@kali:~#
root@kali:~# ls -l traffic/extracted building.jpg
-rw-r--r-- 1 root root 68661 Nov 10 22:27 traffic/extracted_building.jpg
root@kali:~# md5sum traffic/extracted_building.jpg
442f74691f6ff9ea59036570eba2e183 traffic/extracted_building.jpg
root@kali:~#
root@kali:~# md5sum /var/www/html/building.jpg
442f74691f6ff9ea59036570eba2e183 /var/www/html/building.jpg
root@kali:~#
```





Size relation between IP packet and TCP payload

10

Verify the length relation between IP packet and TCP payload

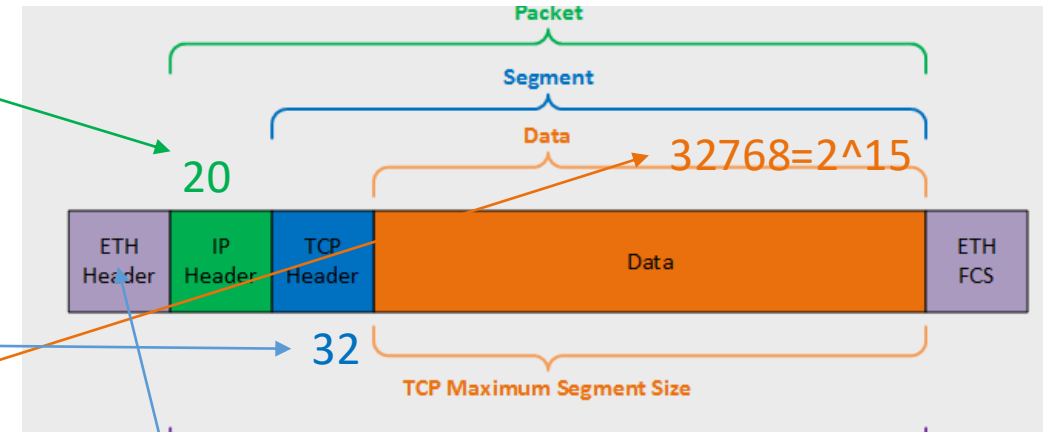
```

10 0.716341907 127.0.0.1 127.0.0.1 TCP 328... 80 → 52054 [ACK] Seq=469 Ack=600 Win=65536 Len=32768 TSval=
11 0.716353387 127.0.0.1 127.0.0.1 TCP 66 52054 → 80 [ACK] Seq=600 Ack=33237 Win=48512 Len=0 TSval=
12 0.716363658 127.0.0.1 127.0.0.1 TCP 328... 80 → 52054 [PSH, ACK] Seq=33237 Ack=600 Win=65536 Len=327
13 0.716368424 127.0.0.1 127.0.0.1 TCP 66 52054 → 80 [ACK] Seq=600 Ack=66005 Win=15744 Len=0 TSval=
14 0.716376090 127.0.0.1 127.0.0.1 HTTP 3480 HTTP/1.1 200 OK (.JPEG .JFIF image)

Frame 10: 32834 bytes on wire (262672 bits), 32834 bytes captured (262672 bits) on interface lo, id 0
Ethernet II, Src: 00:00:00_00:00:00 (00:00:00:00:00:00), Dst: 00:00:00_00:00:00 (00:00:00:00:00:00)
Internet Protocol Version 4, Src: 127.0.0.1, Dst: 127.0.0.1
  0100 .... = Version: 4
  .... 0101 = Header Length: 20 bytes (5)
  Differentiated Services Field: 0x00 (DSCP: CS0, ECN: Not-ECT)
  Total Length: 32820
  Identification: 0xab19 (43801)

Transmission Control Protocol, Src Port: 80, Dst Port: 52054, Seq: 469, Ack: 600, Len: 32768
  Source Port: 80
  Destination Port: 52054
  [Stream index: 0]
  [TCP Segment Len: 32768]
  Sequence number: 469 (relative sequence number)
  Sequence number (raw): 1720159393
  [Next sequence number: 33237 (relative sequence number)]
  Acknowledgment number: 600 (relative ack number)
  Acknowledgment number (raw): 1950579563
  1000 .... = Header Length: 32 bytes (8)
  Flags: 0x010 (ACK)
  Window size value: 512
  [Calculated window size: 65536]
  [Window size scaling factor: 128]
  Checksum: 0x7e29 [unverified]
  [Checksum Status: Unverified]
  Urgent pointer: 0
  Options: (12 bytes), No-Operation (NOP), No-Operation (NOP), Timestamps
  [SEQ/ACK analysis]
  [Timestamps]
  TCP payload (32768 bytes)

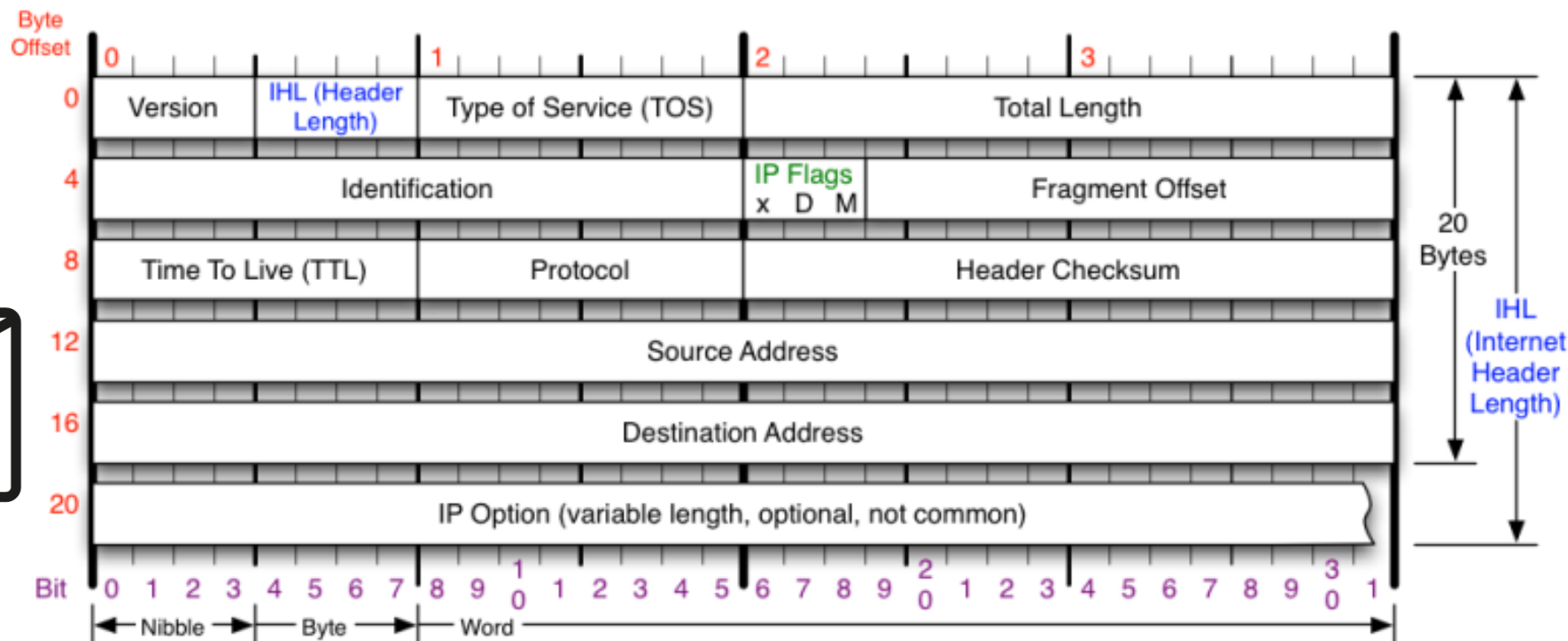
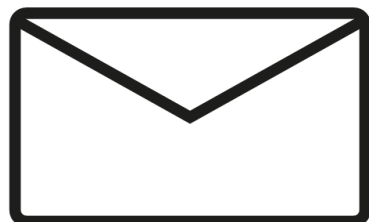
```



TCP segment length needs to be calculated



IP Header



Version

Version of IP Protocol. 4 and 6 are valid. This diagram represents version 4 structure only.

Header Length

Number of 32-bit words in TCP header, minimum value of 5. Multiply by 4 to get byte count.

Protocol

IP Protocol ID. Including (but not limited to):

1 ICMP	17 UDP	57 SKIP
2 IGMP	47 GRE	88 EIGRP
6 TCP	50 ESP	89 OSPF
9 IGRP	51 AH	115 L2TP

Total Length

Total length of IP datagram, or IP fragment if fragmented. Measured in Bytes.

Fragment Offset

Fragment offset from start of IP datagram. Measured in 8 byte (2 words, 64 bits) increments. If IP datagram is fragmented, fragment size (Total Length) must be a multiple of 8 bytes.

Header Checksum

Checksum of entire IP header

IP Flags

x D M

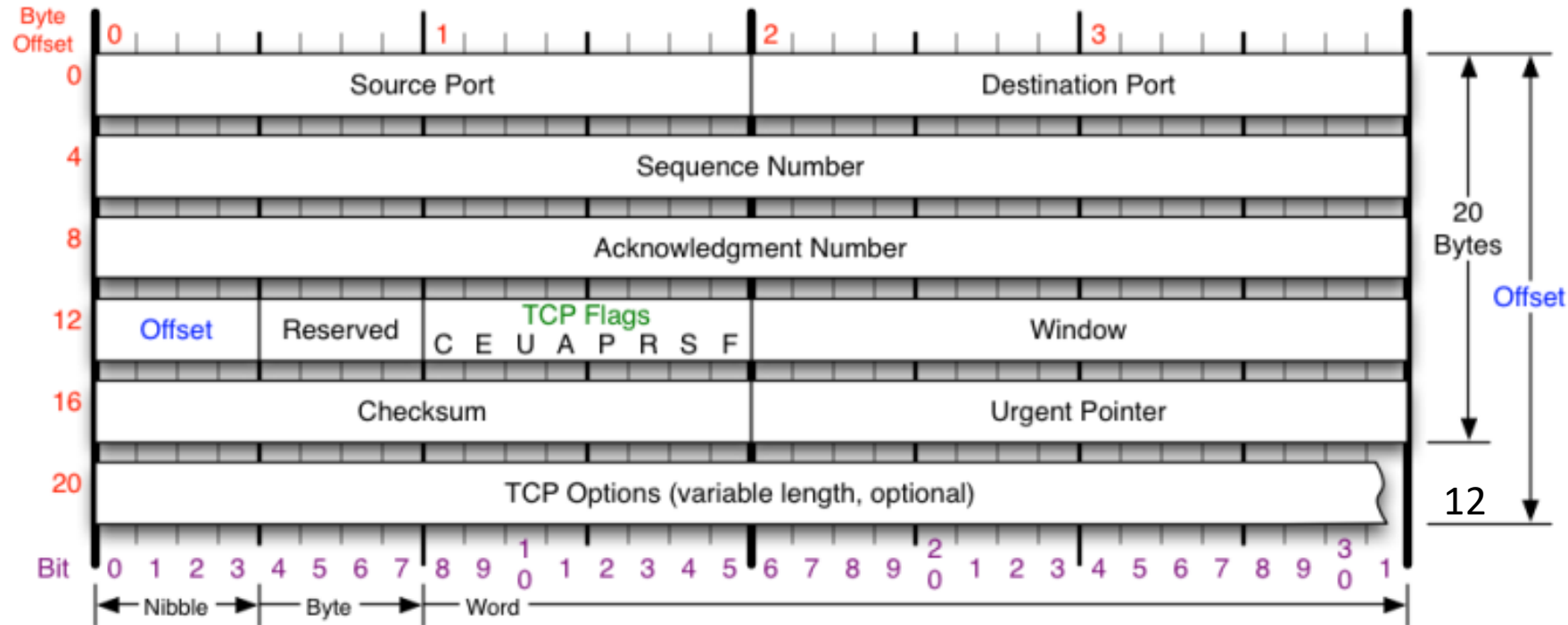
x 0x80 reserved (evil bit)
D 0x40 Do Not Fragment
M 0x20 More Fragments follow

RFC 791

Please refer to RFC 791 for the complete Internet Protocol (IP) Specification.



Tcp Header



TCP Flags

C E U A P R S F

Congestion Window

- C 0x80 Reduced (CWR)
- E 0x40 ECN Echo (ECE)
- U 0x20 Urgent
- A 0x10 Ack
- P 0x08 Push
- R 0x04 Reset
- S 0x02 Syn
- F 0x01 Fin

Congestion Notification

ECN (Explicit Congestion Notification). See RFC 3168 for full details, valid states below.

Packet State	DSB	ECN bits
Syn	0 0	1 1
Syn-Ack	0 0	0 1
Ack	0 1	0 0
No Congestion	0 1	0 0
No Congestion	1 0	0 0
Congestion	1 1	0 0
Receiver Response	1 1	0 1
Sender Response	1 1	1 1

TCP Options

- 0 End of Options List
- 1 No Operation (NOP, Pad)
- 2 Maximum segment size
- 3 Window Scale
- 4 Selective ACK ok
- 8 Timestamp

Checksum

Checksum of entire TCP segment and pseudo header (parts of IP header)

Offset

Number of 32-bit words in TCP header, minimum value of 5. Multiply by 4 to get byte count.

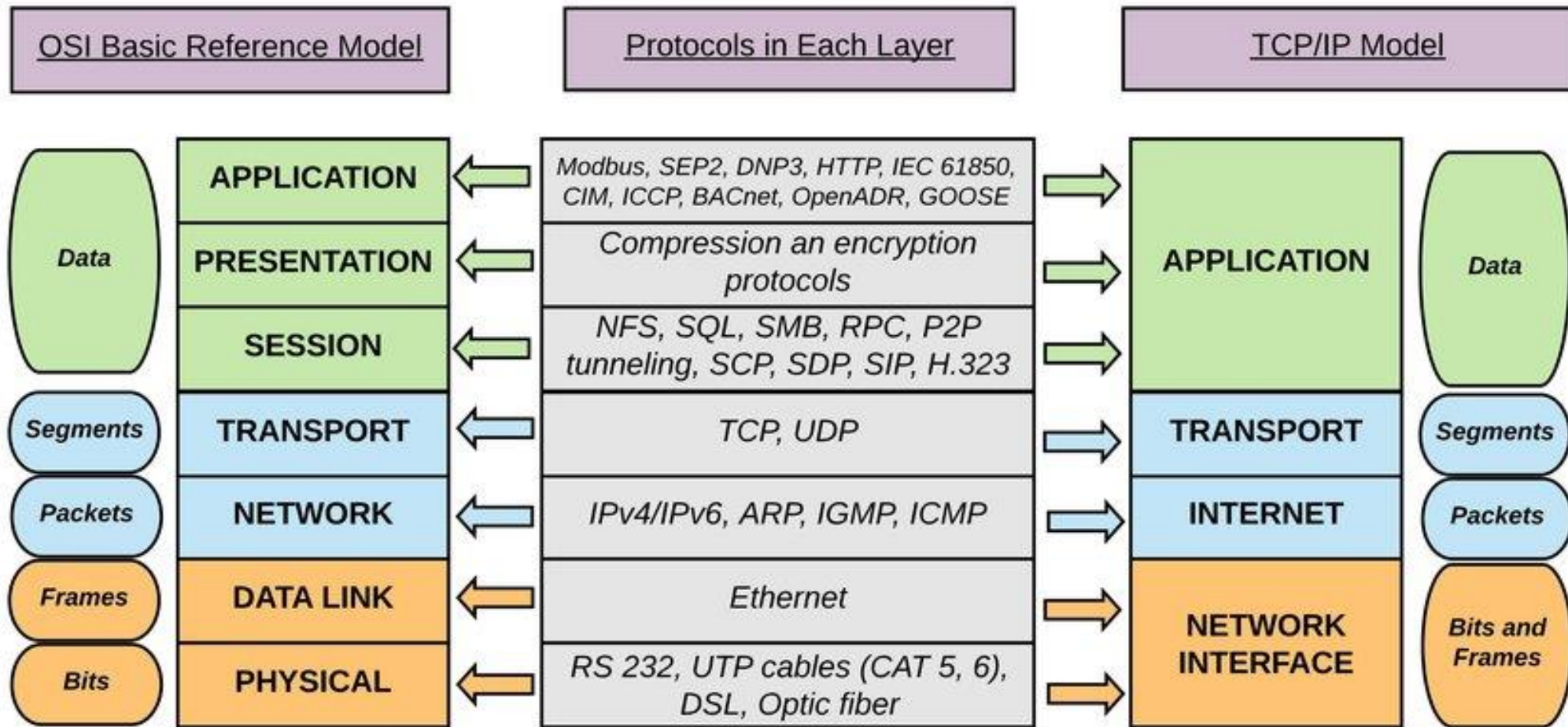
RFC 793

Please refer to RFC 793 for the complete Transmission Control Protocol (TCP) Specification.

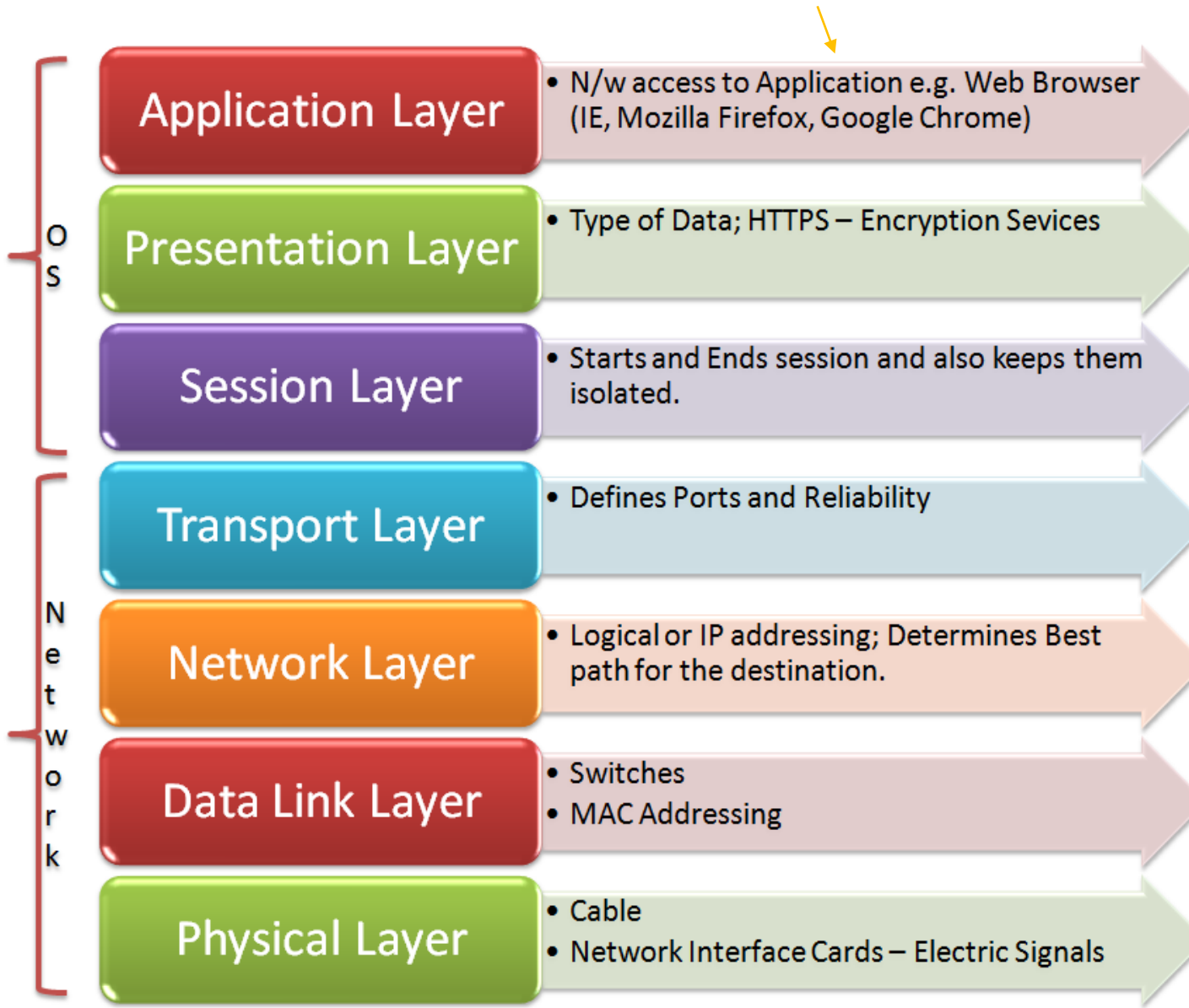


OSI, TCP/IP, security attacks, and forensic evidence

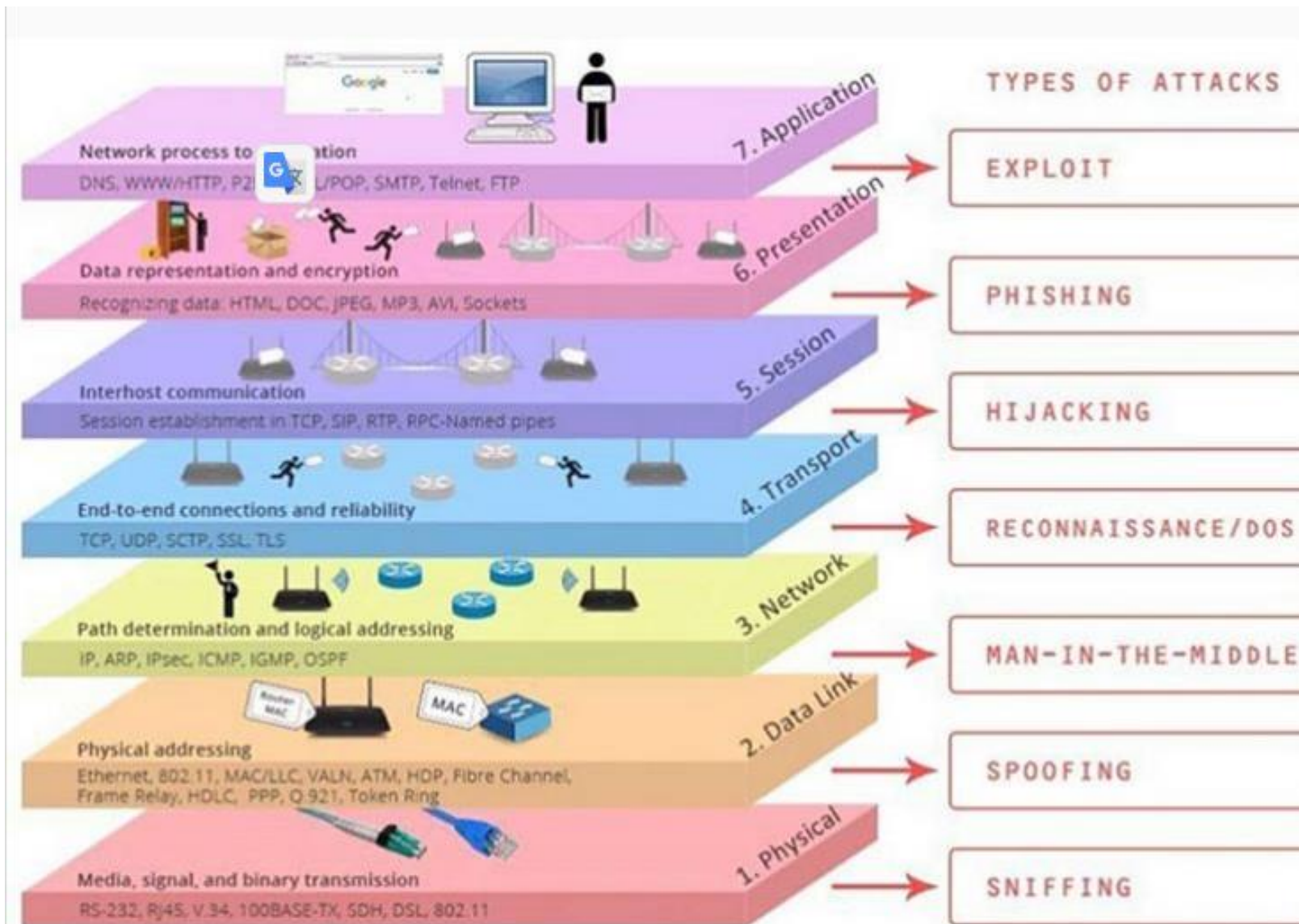




Evidence you should collect



File, images, Software installed, etc.



Assignment



Requirements

- Using curl to sent request to the local web server
 - Capture the http requests and responses in image.log
 - Why can't you see the TCP segments?
- Using Chrome to sent requests from Win 10 and save the log file in image3.log.
 - insert any one photo to .html
 - Extract the photo



Make sure:

- service apache2 start
- has traffic folder

```
/bin/bash 85x8
root@kali:~# rm -r ~/.cache/mozilla/firefox/*default* 1
rm: cannot remove '/root/.cache/mozilla/firefox/*default*': No s
root@kali:~# wireshark -i lo -k -w ~/traffic/image.log 2
Warning: Ignoring XDG_SESSION_TYPE=wayland on Gnome. Use QT_QPA_
un on Wayland anyway.
^C 4
root@kali:~#

/bin/bash 85x14
root@kali:~# curl 127.0.0.1/image.html 3
<!DOCTYPE html>
<html>
<body>

<h1>My First Heading</h1>

<p>My first paragraph.</p>



</body>
</html>
root@kali:~#
```



root@kali:~# **wireshark ~/traffic/image.log**

image.log

File Edit View Go Capture Analyze Statistics Telephony Wireless Tools Help

Apply a display filter ... <Ctrl-/>

No.	Time	Source	Destination	Protocol	Length	Info
1	0.000000000	127.0.0.1	127.0.0.1	TCP	74	51740 → 80 [SYN] Seq=0 Win=65495 Len=0 MSS=65495 SACK_P
2	0.000009452	127.0.0.1	127.0.0.1	TCP	74	80 → 51740 [SYN, ACK] Seq=0 Ack=1 Win=65483 Len=0 MSS=6
3	0.000015344	127.0.0.1	127.0.0.1	TCP	66	51740 → 80 [ACK] Seq=1 Ack=1 Win=65536 Len=0 TSval=1101
4	0.000036057	127.0.0.1	127.0.0.1	HTTP	149	GET /image.html HTTP/1.1
5	0.000042923	127.0.0.1	127.0.0.1	TCP	66	80 → 51740 [ACK] Seq=1 Ack=84 Win=65408 Len=0 TSval=1101
6	0.000289492	127.0.0.1	127.0.0.1	HTTP	456	HTTP/1.1 200 OK (text/html)
7	0.000298548	127.0.0.1	127.0.0.1	TCP	66	51740 → 80 [ACK] Seq=84 Ack=391 Win=65152 Len=0 TSval=1101
8	0.000374524	127.0.0.1	127.0.0.1	TCP	66	51740 → 80 [FIN, ACK] Seq=84 Ack=391 Win=65536 Len=0 TSval=1101
9	0.000413720	127.0.0.1	127.0.0.1	TCP	66	80 → 51740 [FIN, ACK] Seq=391 Ack=85 Win=65536 Len=0 TSval=1101
10	0.000417236	127.0.0.1	127.0.0.1	TCP	66	51740 → 80 [ACK] Seq=85 Ack=392 Win=65536 Len=0 TSval=1101

handshake

http request

http response

teardown

Frame 1: 74 bytes on wire (592 bits), 74 bytes captured (592 bits) on interface lo, id 0

Ethernet II, Src: 00:00:00_00:00:00 (00:00:00:00:00:00), Dst: 00:00:00_00:00:00 (00:00:00:00:00:00)

Internet Protocol Version 4, Src: 127.0.0.1, Dst: 127.0.0.1

Transmission Control Protocol, Src Port: 51740, Dst Port: 80, Seq: 0, Len: 0



```
/bin/bash
/bin/bash 61x6
root@kali:~#
root@kali:~# wireshark -i eth0 -k -w ~/traffic/image3.log 1
Warning: Ignoring XDG_SESSION_TYPE=wayland on Gnome. Use QT_Q
PA_PLATFORM=wayland to run on Wayland anyway.
^C
root@kali:~# 3. Control + C
/bin/bash 61x10
root@kali:~# ifconfig
eth0: flags=4163<UP,BROADCAST,RUNNING,MULTICAST> mtu 1500
    inet 10.0.2.14 netmask 255.255.255.0 broadcast 10.0
    .2.255
    inet6 fe80::a00:27ff:fe7d:bd8b prefixlen 64 scopeid
    0x20<link>
    ether 08:00:27:7d:bd:8b txqueuelen 1000 (Ethernet)
    RX packets 2410 bytes 2180169 (2.0 MiB)
    RX errors 0 dropped 0 overruns 0 frame 0
    TX packets 1953 bytes 220654 (215.4 KiB)
    TX errors 0 dropped 0 overruns 0 carrier 0 collisi
```

